Discrete Mathematics and Logic

Build precise reasoning with propositions, sets, and relations.

Beginner lesson · Estimated reading time: 8–12 minutes

Introduction

Discrete mathematics studies structures made of distinct objects. Logic helps express and test claims; sets organize collections; relations describe connections between objects. These ideas underpin algorithms, databases, digital circuits, and formal verification.

Learning objectives

- Distinguish propositions from questions or commands.
- Evaluate NOT, AND, OR, and implication.
- Use truth tables to test logical statements.
- Apply basic set notation and understand relations.

1. Propositions and truth values

A proposition is a declarative statement that is either true or false in a given interpretation. “7 is prime” is true; “10 < 3” is false. “Close the door” is a command, not a proposition. A variable such as P can stand for a proposition.

Negation NOT P reverses the truth value of P. If P is true, ¬P is false; if P is false, ¬P is true.

2. Logical connectives

Conjunction $P \wedge Q$ (AND) is true only when both P and Q are true. Disjunction $P \vee Q$ (OR) is true when at least one is true, including when both are true. Negation ¬P reverses P.

Implication $P \rightarrow Q$ is false only when P is true and Q is false. Read it as “if P, then Q.” If P is false, the implication is true in classical logic, regardless of Q.

3. Truth tables

A truth table lists every possible truth-value combination. For two propositions P and Q, there are $2^2 = 4$ rows.

P	Q	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	F	T	T

F	F	F	F	T

To compare two expressions, evaluate both columns row by row. If they match on every row, the expressions are logically equivalent.

4. Sets and set operations

A set is a collection of distinct elements. If A = {1, 2, 3} and B = {3, 4}, then A ∪ B = {1, 2, 3, 4} (union), A ∩ B = {3} (intersection), and A \ B = {1, 2} (difference). The order of elements does not matter, and repeated elements do not create new members.

Use $x \in A$ to mean "x is an element of A" and $A \subseteq B$ to mean every element of A is also in B. The empty set ∅ contains no elements.

5. Relations and applications

A relation from A to B is a set of ordered pairs drawn from A × B. For example, "is less than" relates some numbers to others. Relations can be represented as tables, pairs, or graphs. In databases, a relation-like table organizes records by fields.

A relation on one set may be reflexive (each element relates to itself), symmetric (aRb implies bRa), or transitive (aRb and bRc imply aRc). Check the definition carefully before assigning a property.

6. Common mistakes

- Treating logical OR as exclusive OR when the context uses inclusive OR.
- Assuming $P \to Q$ is the same as $Q \to$

P.

- Confusing an element $x \in A$ with a subset $A \subseteq B$.
- Counting duplicate set elements more than once.
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Declaring a relation symmetric or transitive from only one example.

Practice exercises

EX1: If P is false and Q is true, evaluate P ∧ Q, P ∨ Q, and P → Q.

Answer: P ∧ Q is false; P ∨ Q is true; P → Q is true because a false antecedent makes the implication true in classical logic.

EX2: Let A = {a, b, c} and B = {b, c, d}. Find A ∩ B and A ∪ B.

Answer: A ∩ B = {b, c}; A ∪ B = {a, b, c, d}.

EX3: Is the implication P → Q always equivalent to Q → P?

Answer: No. For P true and Q false, P → Q is false while Q → P is true.